



FORT STREET HIGH SCHOOL

2010

HIGHER SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 4: TRIAL HSC

Mathematics Extension 2

TIME ALLOWED: 3 HOURS
(PLUS 5 MINUTES READING TIME)

Outcomes Assessed	Questions	Marks
Determines the important features of graphs of a wide variety of functions, including conic sections	2, 4	
Applies appropriate algebraic techniques to complex numbers and polynomials	1, 3	
Applies further techniques of integration, such as slicing and cylindrical shells, integration by parts and recurrence formulae, to problems	5, 6	
Synthesises mathematical solutions to harder problems and communicates them in an appropriate form, resisted motion	8, 7	

Question	1	2	3	4	5	6	7	8	Total	%
Marks	/15	/15	/15	/15	/15	/15	/15	/15	/120	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board - approved calculators may be used
- Each new question is to be started in a new booklet

Names: _____

Teacher:
Please tick

☐ Mr Bayas

☐ Mr Fraser

☐ Mr Hayes



Question 1 (15 Marks) Start a new booklet

Marks

- a) Let $z = 5 - 6i$ and $w = 3 + 4i$. Express the following in the form $a + ib$ where a and b are real numbers.

(i) z^2

1

(ii) $\frac{z}{w}$

2

- b) (i) Express $w = 8 + 8i$ in modulus-argument form

1

- (ii) Hence, or otherwise find all numbers z such that $z^5 = 8 + 8i$ giving your answer in modulus-argument form.

3

- c) Sketch the region in the Argand diagram defined by $|z - 2 + i| < 3$ and $-\frac{\pi}{3} \leq \arg(z - 2 + i) \leq \frac{\pi}{3}$

3

Indicate whether corner points are included or excluded. You do not need to find coordinates of the corner points or intercepts.

- d) Find $\sqrt{1+i}$ in the form $a + ib$ where a and b are real numbers. Hence find an exact value for $\tan\left(\frac{\pi}{8}\right)$.

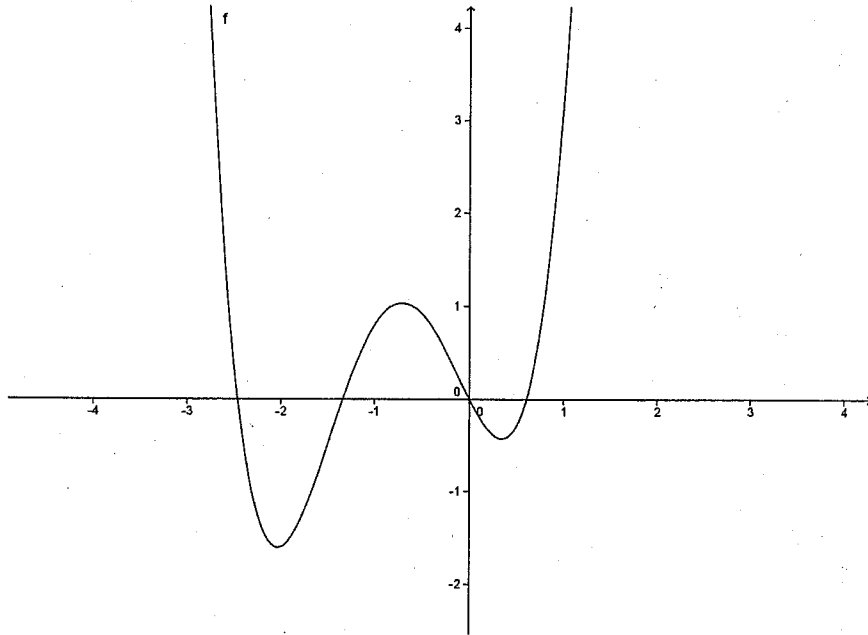
3

- e) Given Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$. The complex number z can be expressed in polar form as $z = re^{i\theta}$ where $r = |z|$ and $\theta = \arg(z)$. Use the polar form of z to find $\ln(z)$ and hence find $\ln(1+i)$ in the form $a + ib$ where a and b are real numbers.

2



Marks

Question 2 (15 Marks) **Start a new booklet**a) The diagram shows the graph of $y = f(x)$ 

Draw separate one third page sketches of the graphs of the following

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y^2 = f(x)$ 2

(iii) $y = 2^{f(x)}$ 2

(iv) $y = f\left(\frac{1}{x}\right)$ 2



Marks

Question 2 continued

- b) Consider the function $f(x) = \ln(2 + 2 \cos(2x))$, $-\pi \leq x \leq 2\pi$
- (i) Show that the function f is even and the curve $y = f(x)$ is concave down for all values of x in its domain, except where its not defined. 3
- (ii) Sketch using a third of a page, the graph of the curve $y = f(x)$. 2
- c) Find the coordinates of the points where the tangent to the curve $x^2 + 2xy + 3y^2 = 18$ is horizontal. 2

End of Question 2

Next question, Question 3 on the next page, page 4

**Question 3 (15 Marks) Start a new booklet**

- a) (i) Prove the theorem
If α is a zero of multiplicity r of the real polynomial equation $P(x) = 0$, then α is a zero of multiplicity $r - 1$ of $P'(x) = 0$.
- (ii) The polynomial equation $3x^5 - ax^2 + b = 0$ has a multiple root.
Show that $8788a^5 = 28125b^3$.
- b) The polynomial $P(z)$ is defined by
$$P(z) = z^4 - 2z^3 - z^2 + 2z + 10$$

Given that $z - 2 + i$ is a factor of $P(z)$, express $P(z)$ as a product of real quadratic factors.
- c) (i) Show that $\cos(P + Q) + \cos(P - Q) = 2\cos P \cos Q$.
- (ii) Let α and β be the roots of the equation $z^2 \sin^2 \phi - z \sin 2\phi + 1 = 0$.
- Show that $\alpha + \beta = 2 \cos \phi \operatorname{cosec} \phi$
 - Show that $\alpha^2 + \beta^2 = 2 \cos 2\phi \operatorname{cosec}^2 \phi$
 - Hence by mathematical induction,
prove that if n is a positive integer then
$$\alpha^n + \beta^n = 2 \cos n\phi \operatorname{cosec}^n \phi$$

Marks

2

3

3

1

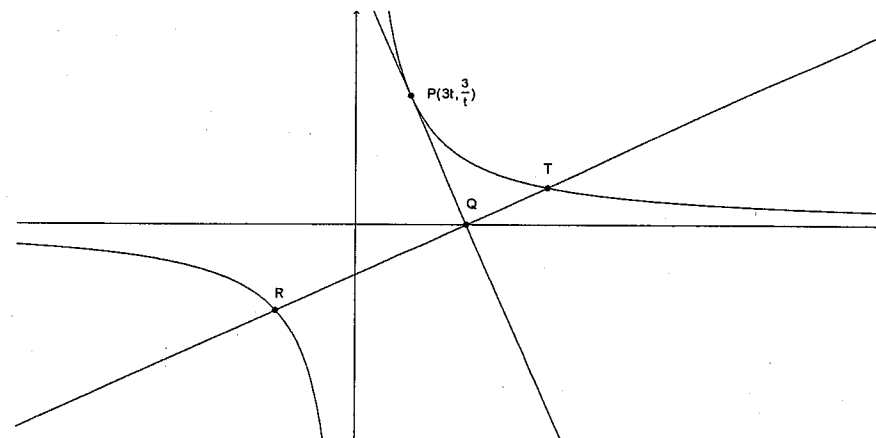
1

1

4

**Question 4 (15 Marks) Start a new booklet****Marks**

- a) $P(3t, \frac{3}{t})$ is a point on the rectangular hyperbola $xy = 9$. The tangent at P cuts the x axis at Q and the line through Q , perpendicular to the tangent at P , cuts the hyperbola at the points R and T as shown



- Show that the equation of the tangent at P is $x + t^2y = 6t$.
 - Show that the line through Q , perpendicular to the tangent at P , has equation $t^2x - y = 6t^3$.
 - If M is the midpoint of RT , show M has coordinates $(3t, -3t^3)$.
 - Find the equation of the locus of M , as P moves on the curve $xy = 9$.
- b) The Hyperbola H has equation $x^2 - 3y^2 = 6$
Show that the equation of the normal to H at $P(2\sqrt{2}, \sqrt{2})$ is $3x + 2y = 8\sqrt{2}$.
- c) The Points $M(a \cos \alpha, b \sin \alpha)$ and $N(-a \sin \alpha, b \cos \alpha)$ lie on the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Find the equations of the tangents at M and N and show these tangents intersect at the point $P(a(\cos \alpha - \sin \alpha), b(\sin \alpha + \cos \alpha))$.

2

3

3

1

2

4

**Question 5 (15 Marks) Start a new booklet****Marks**

- a) Evaluate correct to 3 decimal places

$$\int_0^1 \frac{e^{2x} dx}{e^{4x} + 1}$$

2

- b) Find

$$\int \frac{dp}{\sqrt{9+8p-p^2}}$$

2

- c) Using the substitution
- $t = \tan \frac{\theta}{2}$
- , find

$$\int \frac{2d\theta}{5-4\sin\theta}$$

3

- d) Find

$$\int \frac{x^5 - 7x^2 + 8}{x^3 - 8} dx$$

4

- e) If
- $I_n = \int_0^{\pi/4} \sec^n x dx$
- for
- $n \geq 0$

(integral from zero to pi over 4 of secx to the power n dx)

4

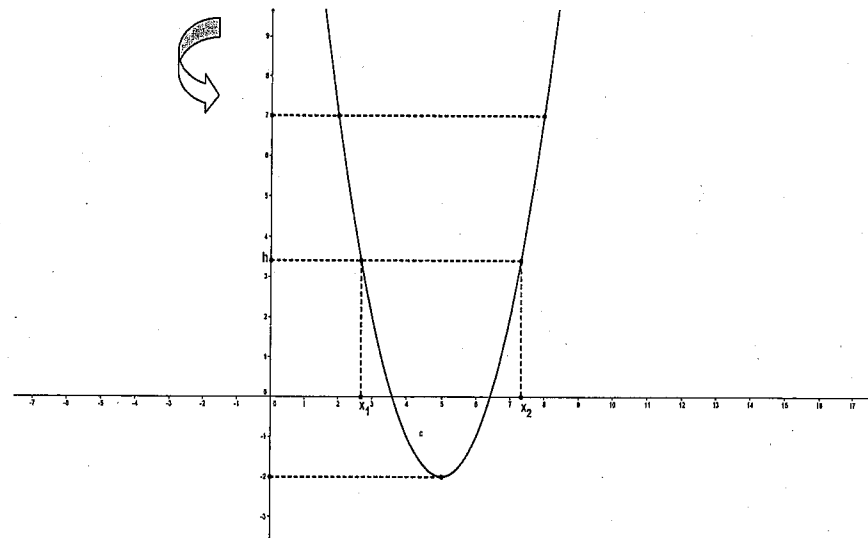
show that

$$I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2} \quad \text{for } n \geq 2$$

$$\text{and deduce } I_6 = \frac{28}{15}$$

**Question 6 (15 Marks) Start a new booklet****Marks**

- a) A flat top parabolic torus is formed by rotating the area inside the parabola
- $y = x^2 - 10x + 23$
- between the lines
- $y = -2$
- and
- $y = 7$
- around the y axis.



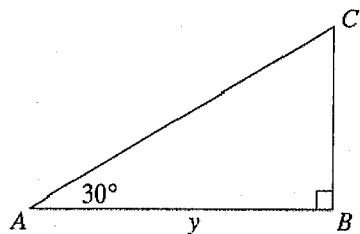
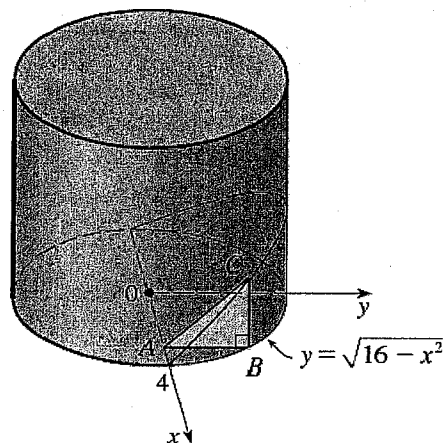
The cross section at $y = h$ where $-2 \leq h \leq 7$, is an annulus. The annulus has inner radius x_1 and outer radius x_2 where x_1 and x_2 are the solutions to $x^2 - 10x + 23 = h$

- (i) Find x_1 and x_2 in terms of h 1
- (ii) Find the area of the cross-section at height h , in terms of h . 2
- (iii) Find the volume of the flat top parabolic torus. 2
Leave answer in exact form.

**Question 6. Continued****Marks**

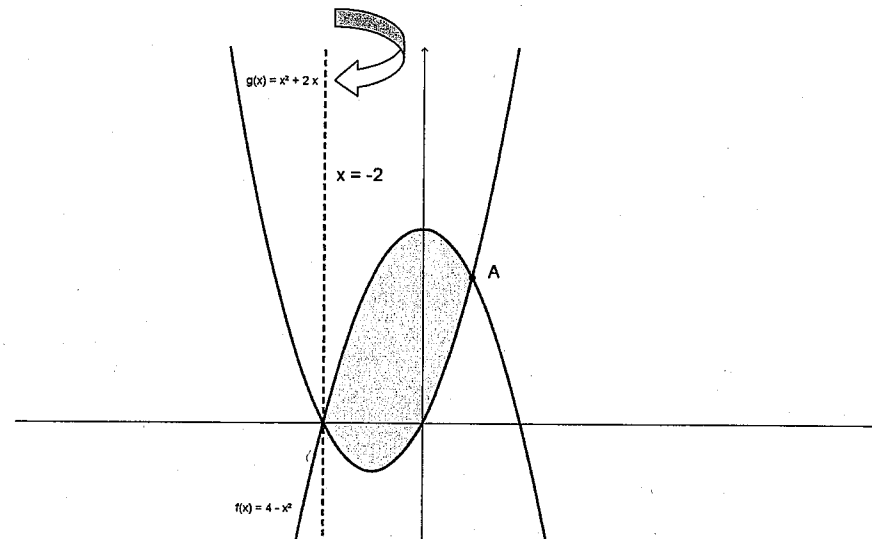
- b) A wedge is cut out of a circular cylinder of radius 4 by two planes. One plane is perpendicular to the axis of the cylinder. The other intersects the first at an angle of 30° along a diameter of the cylinder.

- (i) Show the cross sectional area is $A(x) = \frac{16-x^2}{2\sqrt{3}}$ 2
- (ii) Hence find the volume of the wedge. 3

**Question 6. continued****Marks**

c)

The lightly shaded region bounded by $y = 4 - x^2$, $y = x^2 + 2x$ is rotated about the line $x = -2$. The point A is the intersection of $y = 4 - x^2$ and $y = x^2 + 2x$ in the first quadrant.



- (i) Find the coordinate of A 1
- (ii) Use the method of cylindrical shells to express the volume of the resulting solid of revolution as an integral. 2
- (iii) Evaluate the integral in part (ii), leave answer in exact form. 2

**Question 7. (15 Marks) Start a new booklet**

Marks

- a) A cannon ball of mass 1 kilogram is projected vertically upward from the origin with an initial speed of 20m/s . The cannon ball is subjected to gravity 10ms^{-2} and air resistance $\frac{v^2}{20}$.

The upward equation of motion is

$$\ddot{y} = -\frac{v^2}{20} - 10$$

- (i) Using $\dot{y} = v \frac{dv}{dy}$ show that while the cannon ball

is rising $v^2 = 600e^{-\frac{y}{10}} - 200$

- (ii) Hence find the maximum height reached by the cannon ball correct to 2 decimal places.

- (iii) Using $\dot{y} = \frac{dv}{dt}$ find how long the cannon ball takes to reach this maximum height correct to 2 decimal places?

- (iv) How fast is the cannon ball travelling when it returns to the origin correct to 2 decimal places?

- b) A cylindrical water tank has a constant cross-sectional area A . Water drains through a hole at the bottom of the tank.

The Volume of water decreases at a rate $\left(-k \text{ times the cube root of } h\right)$
 $\frac{dV}{dt} = -k\sqrt[3]{h}$ Where k is a positive constant and h is the depth of water.

Initially the tank is full and it takes T seconds to drain. Thus $h = h_0$ when $t = 0$

And $h = 0$ when $t = T$.

- (i) Show that $\frac{dh}{dt} = -\frac{k}{A}\sqrt[3]{h}$

- (ii) By considering the equation for $\frac{dt}{dh}$ or otherwise

$$\text{Show } h^2 = h_0^2 \left(1 - \frac{t}{T}\right)^3.$$

- (iii) Suppose it takes 12 seconds for half the water to drain. How long does it take to empty the full tank?

to nearest second

**Question 8. (15 Marks) Start a new booklet**

Marks

- a) Let α be a real number and suppose z is a complex number such that

$$z + \frac{1}{z} = 2\cos \alpha$$

- (i) By reducing the above equation to a quadratic equation in z , solve for z and use de Moivre's theorem to show that

$$z^n + \frac{1}{z^n} = 2\cos n\alpha.$$

- (ii) Let $w = z + \frac{1}{z}$. Prove that

$$w^3 + w^2 - 2w - 2 = z + \frac{1}{z} + \left(z^2 + \frac{1}{z^2}\right) + \left(z^3 + \frac{1}{z^3}\right).$$

- (iii) Hence, or otherwise, find all solutions of $\cos \alpha + \cos 2\alpha + \cos 3\alpha = 0$, in the range $0 \leq \alpha \leq 2\pi$.

- b) Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$,

Hence evaluate $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$.

- c) The area A of the surface of revolution generated by rotating a smooth arc $y = f(x)$, $a \leq x \leq b$ around the x axis, is given by the integral formula

$$A = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx$$

Rotate the circle $x^2 + y^2 = r^2$ around the x axis and show that the surface Area of the sphere generated is $4\pi r^2$.



End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 [15 Marks]

a) i) $z^2 = (5-6i)(5-6i) = 25 - 60i + 36i^2 = -11 - 60i$ [1]

ii) $\frac{z}{w} = \frac{5-6i}{3-4i} \times \frac{3+4i}{3+4i} = \frac{15+20i-18i-24i^2}{9+16} = \frac{39+2i}{25}$ [1]

b) i) $8+8i = \sqrt{8^2+8^2} \operatorname{cis}(\tan^{-1}(1)) = 8\sqrt{2} \operatorname{cis} \frac{\pi}{4}$
 $|w| = 8\sqrt{2}, \arg(w) = \tan^{-1}(1) = \frac{\pi}{4}$ [1]

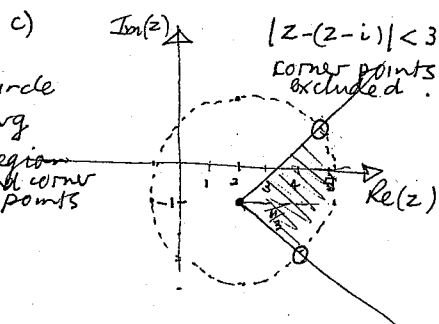
ii) $z^5 = 8+8i = 8\sqrt{2} \operatorname{cis}(\frac{\pi}{4} + 2k\pi)$ $k=0,1,2,3,4$ unique
 $z = (8\sqrt{2})^{\frac{1}{5}} \operatorname{cis}(\frac{1}{5}(\frac{\pi}{4} + 2k\pi))$

[1] $2k\pi$

[1] $2^{\frac{7}{10}}$

[1] all correct arguments.

$z_0 = 2^{\frac{7}{10}} \operatorname{cis} \frac{\pi}{20}, z_1 = 2^{\frac{7}{10}} \operatorname{cis} \frac{9\pi}{20}, z_2 = 2^{\frac{7}{10}} \operatorname{cis} \frac{17\pi}{20}$
 $z_3 = 2^{\frac{7}{10}} \operatorname{cis} \frac{5\pi}{4}, z_4 = 2^{\frac{7}{10}} \operatorname{cis} \frac{33\pi}{20}$



d) $\sqrt{1+i} = a+ib$ $a, b > 0$

① $\therefore a^2 - b^2 = 1$ $2ab = 1 \therefore b = \frac{1}{2a}$

$\therefore a^2 - (\frac{1}{2a})^2 = 1 \Rightarrow 4a^4 - 1 = 4a^2$

$\therefore 4a^4 - 4a^2 - 1 = 0$ let $p = a^2$

$\therefore 4p^2 - 4p - 1 = 0$

$\therefore p = \frac{4 \pm \sqrt{16+16}}{8} = \frac{1 \pm \sqrt{2}}{2}$ $p > 0$

$\therefore p = \frac{1+\sqrt{2}}{2} \therefore a = \sqrt{\frac{1+\sqrt{2}}{2}}, a > 0$

Now $a^2 - b^2 = 1$ from ①

$\therefore \frac{1+\sqrt{2}}{2} - b^2 = 1 \therefore b^2 = \frac{\sqrt{2}-1}{2}$

$\therefore a = \sqrt{\frac{1+\sqrt{2}}{2}}, b = \sqrt{\frac{\sqrt{2}-1}{2}}$ [1]

$\sqrt{1+i}$ has arg of $\frac{\pi}{8}$ [1]

$\therefore \tan \frac{\pi}{8} = \frac{b}{a} = \frac{\sqrt{\frac{\sqrt{2}-1}{2}}}{\sqrt{\frac{1+\sqrt{2}}{2}}} = \underline{\underline{\sqrt{2}-1}}$

Simplify

$\sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}} = \sqrt{2}-1$

e) Euler

$e^{i\theta} = \operatorname{cis} \theta$

$z = re^{i\theta}$

if $z = 1+i$ $|z| = \sqrt{2}$ $\arg z = \frac{\pi}{4}$

[1] $\therefore z = \sqrt{2} \operatorname{cis} \frac{\pi}{4} = \sqrt{2} e^{i\frac{\pi}{4}}$

$\ln z = \ln \sqrt{2} e^{i\frac{\pi}{4}}$

$= \ln \sqrt{2} + \ln e^{i\frac{\pi}{4}}$

$= \frac{1}{2} \ln 2 + i \frac{\pi}{4}$

$\therefore a = \ln \sqrt{2}$ or $\frac{1}{2} \ln 2$

$b = \frac{\pi}{4}$

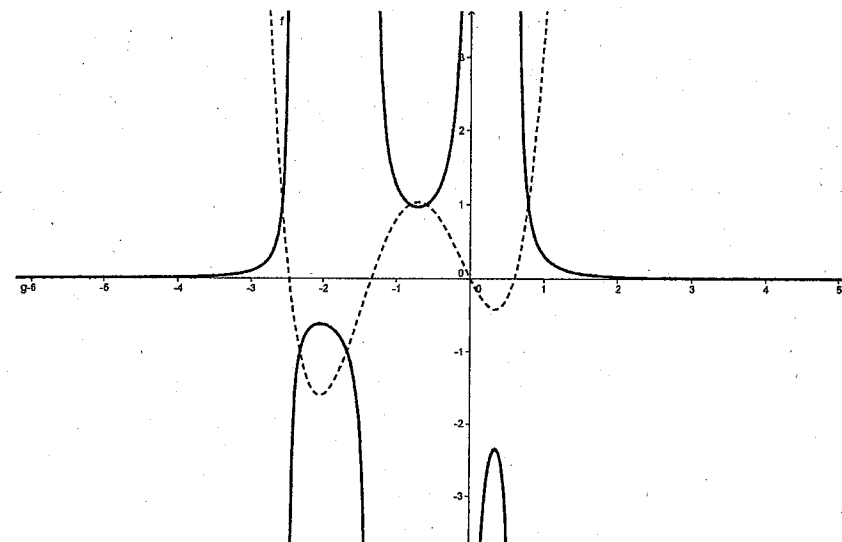
[1]

ie $\ln z = \frac{1}{2} \ln 2 + \frac{\pi}{4} i$

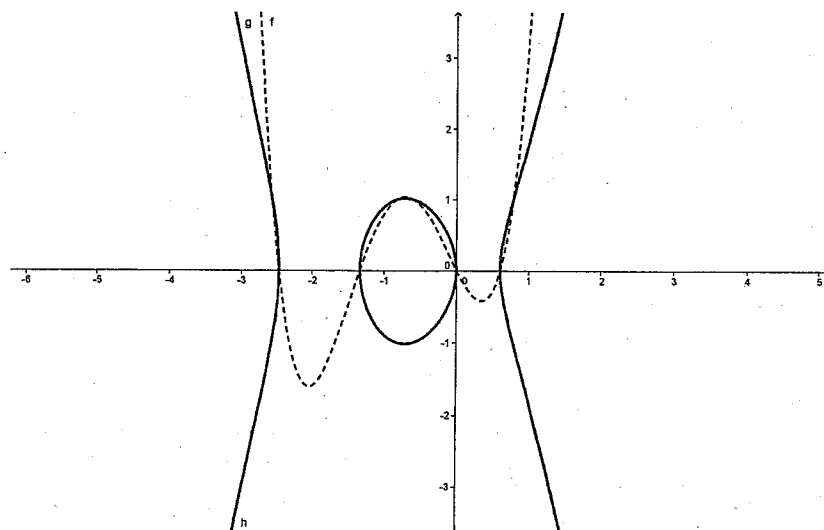
where $z = 1+i$

Q2 a)

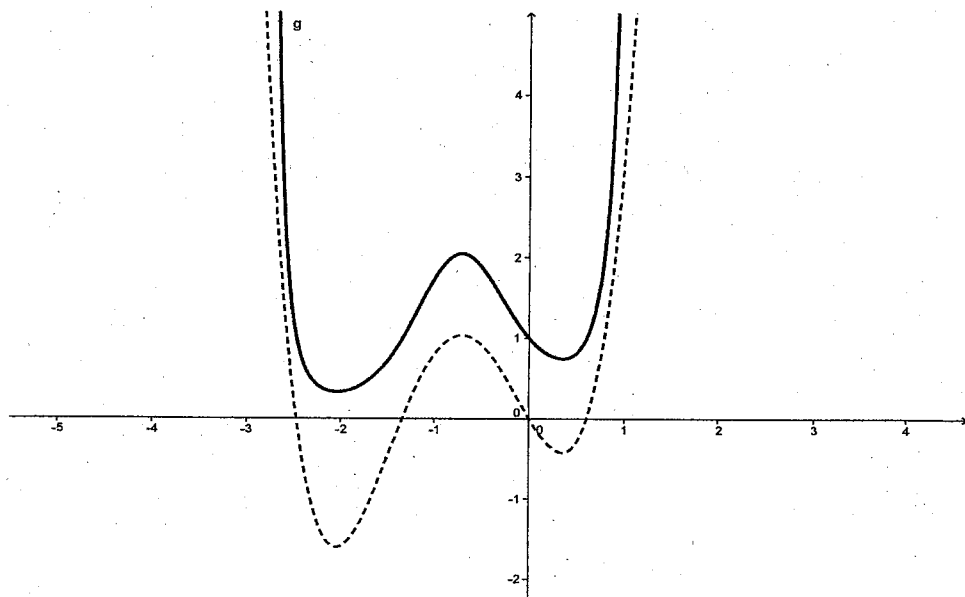
(i)



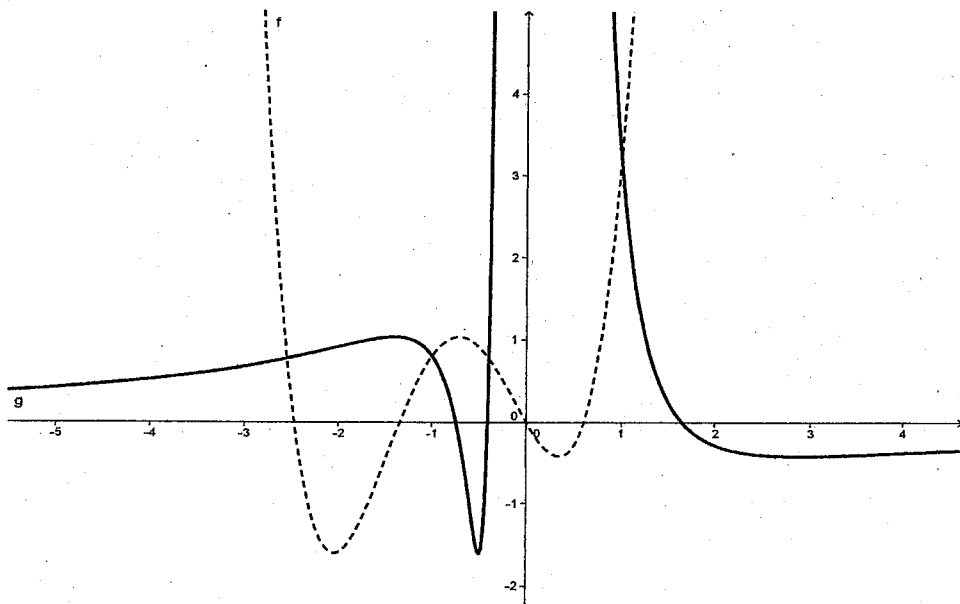
(ii)



(iii)



iv)

Question 2

b) $f(x) = \ln(2 + 2\cos(2x))$

$$\begin{aligned} \text{(i)} \quad f(-x) &= \ln(2 + 2\cos(-2x)) \text{ as } \cos(-2x) = \cos 2x \\ &= \ln(2 + 2\cos(2x)) \\ &= f(x) \therefore f(x) \text{ is even} \end{aligned}$$

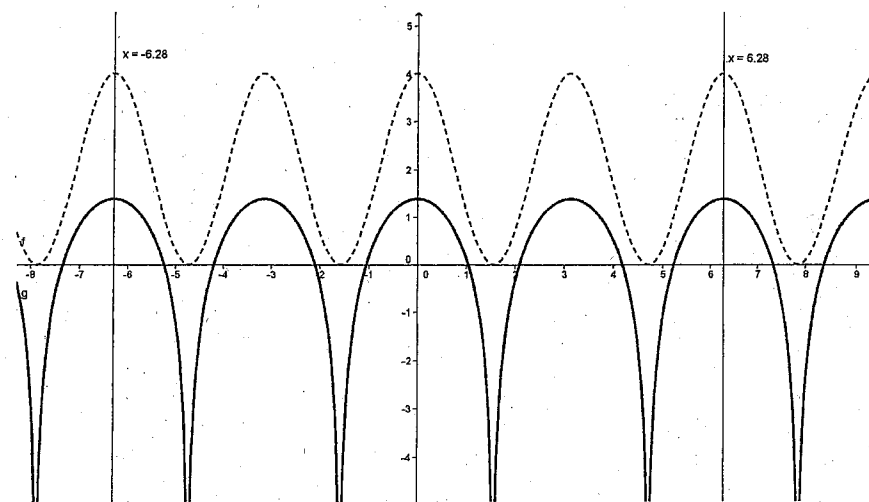
[1]

$$f'(x) = \frac{-4\sin 2x}{2 + 2\cos 2x} = \frac{-2\sin 2x}{1 + \cos 2x}$$

$$\begin{aligned} f''(x) &= \frac{(1 + \cos 2x) \cdot (-4\cos 2x) + (2\sin 2x) \cdot (-2\sin 2x)}{(1 + \cos 2x)^2} \\ &= \frac{-4\cos 2x - 4\cos^2 2x - 4\sin^2 2x}{(1 + \cos 2x)^2} \quad -1 \leq \cos 2x \leq 1 \\ &= \frac{-4(1 + \cos 2x)}{(1 + \cos 2x)^2} = \frac{-4}{1 + \cos 2x} \end{aligned}$$

[2] $\therefore f'(x) < 0$ except where $\cos 2x = -1$ where not defined.

(ii) Sketch [2]



Q2 c) $x^2 + 2xy + 3y^2 = 18$

$$\therefore 2x + 2x \frac{dy}{dx} + 2y + 6y^2 \frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = \frac{-(2x + 2y)}{2x + 6y^2} \quad [1] \text{ } \frac{dy}{dx} \text{ implicit correct}$$

If tangent horizontal $\frac{dy}{dx} = 0 \quad \therefore 2x + 2y = 0 \quad \therefore x = -y$

Sub into original eqn

$$\therefore y^2 - 2y^2 + 3y^2 = 18 \quad \therefore 2y^2 = 18 \quad y^2 = 9 \quad \therefore y = \pm 3$$

$\therefore$ points $(3, -3)$ and $(-3, 3)$

[1] correct points

Question 3 (15 Marks)

a) i) $P(x) = (x-\alpha)^r Q(x) \therefore P'(x) = r(x-\alpha)^{r-1} Q(x) + (x-\alpha)^r Q'(x)$
 $\therefore P'(x) = (x-\alpha)^{r-1} [rQ(x) + (x-\alpha)Q'(x)]$
 $\therefore \alpha$ is a root of multiplicity $r-1$ of $P'(x) = 0$.

ii) $P(x) = 3x^5 - ax^2 + b = 0$
 $\therefore P'(x) = 15x^4 - 2ax = 0 \therefore x(15x^3 - 2a) = 0$
 $\therefore x=0$ or $15x^3 = 2a \therefore x = (\frac{2a}{15})^{\frac{1}{3}}$

Sub into $P(x) = 3(\frac{2a}{15})^{\frac{5}{3}} - a(\frac{2a}{15})^{\frac{2}{3}} + b = 0$
 $3a^{\frac{5}{3}}(\frac{2}{15})^{\frac{5}{3}} - a^{\frac{2}{3}}(\frac{2}{15})^{\frac{2}{3}} + b = 0$
 $a^{\frac{5}{3}}[3(\frac{2}{15})^{\frac{5}{3}} - (\frac{2}{15})^{\frac{2}{3}}] = -b$
 $a^{\frac{5}{3}}[3(\frac{2}{15})^{\frac{2}{3}}[\frac{2}{15} - \frac{1}{3}]] = -b$
 $a^{\frac{5}{3}}[3(\frac{2}{15})^{\frac{2}{3}} \cdot \frac{-1}{15}] = -b$ cube both sides
 $a^5 \cdot 3^3 (\frac{2}{15})^2 (\frac{-1}{15})^3 = (-b)^3$
 $\therefore a^5 \cdot 3^3 \cdot 2^2 \cdot (-1)^3 = 15^2 \cdot 15^3 \cdot -b^3$
 $-237276a^5 = -759375b^3$
 $\therefore 8788a^5 = 28125b^3 \Rightarrow 12a^5 = 3125b^3$

b) $z-2i$ factor $\therefore z-(2-i) \rightarrow 2-i$ zero, Real coeff.
 $\therefore z+i$ is also a zero, Hence $(z-(2-i))(z-(2+i))$ is a factor

ii $z^2 - 4z + 5$ is a factor

$$\begin{array}{r} z^2 + 2z + 2 \\ z^2 - 4z + 5 \overline{) z^4 - 2z^3 - z^2 + 2z + 10} \\ \underline{z^4 - 4z^3 + 5z^2 -} \\ 2z^3 - 6z^2 + 2z + 10 \\ \underline{2z^3 - 8z^2 + 10z -} \\ 2z^2 - 8z + 10 \\ \underline{2z^2 - 8z + 10 -} \\ 0 \end{array}$$

$\therefore P(z) = (z^2 - 4z + 5)(z^2 + 2z + 2)$
 (product of real quadratic factors)

Question 3 (cont)

i) $\cos(P+Q) + \cos(P-Q)$
 $= \cos P \cos Q - \sin P \sin Q + \cos P \cos Q + \sin P \sin Q$
 $= 2 \cos P \cos Q$

ii) $z^2 \sin^2 \phi - z \sin 2\phi + 1 = 0 \quad \alpha\beta = \frac{1}{\sin^2 \phi} = \operatorname{cosec}^2 \phi$
 1. $\alpha + \beta = \frac{\sin 2\phi}{\sin^2 \phi} = \frac{2 \sin \phi \cos \phi}{\sin^2 \phi} = \frac{2 \cos \phi}{\sin \phi}$
 $= 2 \cos \phi \operatorname{cosec} \phi$

2. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (2 \cos \phi \operatorname{cosec} \phi)^2 - 2 \operatorname{cosec}^2 \phi$
 $= (2 \cos^2 \phi - 1) \operatorname{cosec}^2 \phi$
 $= \cos 2\phi \operatorname{cosec}^2 \phi$

3. from 1. and 2. the formula is true for $n=1$ and $n=2$

Assume true for $n=k, k-1$ (for all n $2 \leq n \leq k$)
 i. $\alpha^k + \beta^k = 2 \cos k\phi \operatorname{cosec}^k \phi, \alpha^{k-1} + \beta^{k-1} = 2 \cos(k-1)\phi \operatorname{cosec}^{k-1} \phi$
 Now prove true for $n=k+1$.

ii $\alpha^{k+1} + \beta^{k+1} = 2 \cos(k+1)\phi \operatorname{cosec}^{k+1} \phi$
 Multiply original equation by z^{k-1} (ii)

$\therefore z^{k+1} \sin^2 \phi - z^k \sin 2\phi + z^{k-1} = 0$
 Sub in α, β
 $\alpha^{k+1} \sin^2 \phi - \alpha^k \sin 2\phi + \alpha^{k-1} = 0$
 $\beta^{k+1} \sin^2 \phi - \beta^k \sin 2\phi + \beta^{k-1} = 0$

add (rearrange)
 $(\alpha^{k+1} + \beta^{k+1}) \sin^2 \phi = (\alpha^k + \beta^k) \sin 2\phi - \alpha^{k-1} - \beta^{k-1}$
 using assumption $= (2 \cos k\phi \operatorname{cosec}^k \phi) \sin 2\phi - (\alpha^{k-1} + \beta^{k-1})$
 divide by $\sin^2 \phi$

$\therefore \alpha^{k+1} + \beta^{k+1} = 2 \cos k\phi \operatorname{cosec}^k \phi \cdot \frac{\sin 2\phi}{\sin^2 \phi} - \frac{2 \cos(k-1)\phi \operatorname{cosec}^{k-1} \phi}{\sin^2 \phi}$
 $= 4 \cos k\phi \operatorname{cosec}^{k+1} \phi \cos \phi - 2 \cos(k-1)\phi \operatorname{cosec}^{k+1} \phi$
 $= 2 \operatorname{cosec}^{k+1} \phi [2 \cos k\phi \cos \phi - \cos(k-1)\phi]$
 $= 2 \operatorname{cosec}^{k+1} \phi [\cos(k\phi + \phi) + \cos(k\phi - \phi) - \cos(k-1)\phi]$
 $= 2 \operatorname{cosec}^{k+1} \phi \cos(k+1)\phi = \text{RHS}$

Hence since formula is true for $n=1, 2$ and with our assumptions on interval $2 \leq n \leq k$ true for $n=k+1$, so by the principle of mathematical induction $\alpha^n + \beta^n = 2 \operatorname{cosec}^n \phi \cos n\phi$ for integers

Question 5 [15 Marks]

a) $\int_0^1 \frac{e^{2x} dx}{e^{4x} + 1}$ let $u = e^{2x}$ $du = 2e^{2x} dx$ $x=0 \Rightarrow u=1$ $x=1 \Rightarrow u=e^2$ [1] Sub + bounds
 $\therefore \frac{1}{2} \int_1^{e^2} \frac{du}{u^2 + 1} = \frac{1}{2} \tan^{-1} u \Big|_1^{e^2} = \frac{1}{2} [\tan^{-1}(e^2) - \tan^{-1}(1)] \div 0.325$ [1] - answer

b) $\int \frac{dp}{\sqrt{9+8p-p^2}} = \int \frac{dp}{\sqrt{-(p^2-8p-9)}} = \int \frac{dp}{\sqrt{-(p-4)^2-25}}$ [1] Comple Square
 $\boxed{SI} = \int \frac{dp}{\sqrt{25-(p-4)^2}} = \sin^{-1}\left(\frac{p-4}{5}\right) + C$ [1] use SI diff

c) $\int \frac{2d\theta}{5-4\sin\theta} = \int \frac{2 \cdot 2dt}{5(1+t^2)-4 \cdot 2t} = \int \frac{4dt}{5t^2-8t+5}$ [1] Sub
 $t = \tan \frac{\theta}{2}$
 $\frac{d\theta}{dt} = \frac{1}{1+t^2}$
 $\therefore d\theta = \frac{2dt}{1+t^2}$ [1] dθ/dt
 $\triangle \begin{matrix} 1+t^2 \\ \hline 1-t^2 \end{matrix}$ [1] Answer

d) $\int \frac{x^5-7x^2+8}{x^3-8} dx = \int x^2 dx + \int \frac{x^2+8}{x^3-8} dx$ [1] division
 $\frac{x^2}{x^3-8} = \frac{x^2}{(x-2)(x^2+2x+4)}$ PF
 $\frac{x^2}{x^3-8} = \frac{A}{x-2} + \frac{Bx+C}{x^2+2x+4}$ [1] PF

PF $\frac{x^2}{(x-2)(x^2+2x+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+2x+4}$
 $\therefore x^2+8 = A(x^2+2x+4) + (Bx+C)(x-2)$
 let $x=2$
 $12 = 12A \therefore A=1$
 Eqn Coeff x^2
 $A+B=1 \therefore B=0$
 Consts.
 $8 = 4A - 2C$
 $4 = -2C \therefore C = -2$ [1]

Question 4 [15 Marks]

a) i) $y = 9x^{-1}$ $y' = -\frac{9}{x^2}$ at $x=3t$, $y' = -\frac{9}{9t^2} = -\frac{1}{t^2}$ [1] slope
 Equation of tangent $y - \frac{1}{t} = -\frac{1}{t^2}(x-3t)$ [1] eqn
 $\therefore t^2 y - 3t = -x + 3t$ ie $x + t^2 y = 6t$

ii) At Q $y=0 \therefore x=6t$ ie Q(6t, 0) [1] Q
 perpendicular slope $m=t^2 \therefore y-0=t^2(x-6t)$ [1] slope
 $\therefore t^2 x - y = 6t^3$ [1] eqn

iii) Solving $t^2 x - y = 6t^3$ and $xy = 9$ for R, T [1] solve
 $\therefore t^2 x - \frac{9}{x} = 6t^3$ ie $t^2 x^2 - 6t^3 x - 9 = 0$ [1] roots
 Sum of roots $x+p = -\frac{b}{a} = \frac{6t^3}{t^2} = 6t$ ie $x+p = 6t$
 Sub $x=3t$ into line $t^2 x - y = 6t^3 \therefore y = t^2 \cdot 3t - 6t^3 = -3t^3$
 $\therefore$ Midpoint (3t, -3t^3) [1] Midpt

iv) Locus of M(3t, -3t^3) $\therefore x=3t \rightarrow t = \frac{x}{3}$ [1] locus
 $y = -3t^3 = -3\left(\frac{x}{3}\right)^3 \therefore y = -\frac{x^3}{9}$ [1] locus

b) $x^2 - 3y^2 = 6 \therefore 2x - 6y \frac{dy}{dx} = 0 \therefore \frac{dy}{dx} = \frac{2x}{6y} = \frac{x}{3y}$
 $\frac{dy}{dx} \Big|_P = \frac{2\sqrt{2}}{3\sqrt{2}} = \frac{2}{3} \therefore$ slope of normal $= -\frac{3}{2}$ as $m_1 m_2 = -1$ [1] slope N
 $\therefore y - \sqrt{2} = -\frac{3}{2}(x - 2\sqrt{2})$ is eqn of normal [1] eqn
 $2y - 2\sqrt{2} = -3x + 6\sqrt{2}$
 ie $3x + 2y = 8\sqrt{2}$

c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \therefore \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \therefore \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$ [1] ①
 At M(a cos α, b sin α) $\therefore m = -\frac{b^2}{a^2} \cdot \frac{a \cos \alpha}{b \sin \alpha} = -\frac{b \cos \alpha}{a \sin \alpha}$ [1] ②
 At N(-a sin α, b cos α) $\therefore m = -\frac{b^2}{a^2} \cdot \frac{-a \sin \alpha}{b \cos \alpha} = \frac{b \sin \alpha}{a \cos \alpha}$
 Eqn of Tangents at M
 $y - b \sin \alpha = -\frac{b \cos \alpha}{a \sin \alpha}(x - a \cos \alpha) \therefore a y \sin \alpha - a b \sin^2 \alpha = -x b \cos \alpha + a b \cos^2 \alpha$
 $\therefore a y \sin \alpha + x b \cos \alpha = a b$ ie $\frac{y \sin \alpha}{b} + \frac{x \cos \alpha}{a} = 1$ ①
 Eqn of Tangents at N
 $y - b \cos \alpha = \frac{b \sin \alpha}{a \cos \alpha}(x + a \sin \alpha) \therefore a y \cos \alpha - a b \cos^2 \alpha = x b \sin \alpha + a b \sin^2 \alpha$
 $\therefore a y \cos \alpha - x b \sin \alpha = a b$ ie $\frac{y \cos \alpha}{b} - \frac{x \sin \alpha}{a} = 1$ ②
 ① $\times \sin \alpha$ $\frac{y \sin^2 \alpha}{b} + \frac{x \cos \alpha \sin \alpha}{a} = \sin \alpha$
 ② $\times \cos \alpha$ $\frac{y \cos^2 \alpha}{b} - \frac{x \sin \alpha \cos \alpha}{a} = \cos \alpha$
 Sub $y = b(\sin \alpha + \cos \alpha)$ into either $\frac{x \cos \alpha}{a} + \frac{b(\sin \alpha + \cos \alpha) \sin \alpha}{b} = 1$ D

$$e) I_n = \int_0^{\frac{\pi}{4}} \sec^n x dx = \int_0^{\frac{\pi}{4}} \sec^{n-2} x \sec^2 x dx$$

$$u = \sec^{n-2} x$$

$$dV = \sec^2 x dx$$

$$[1] u = \sec^{n-2} x$$

$$du = (n-2) \sec^{n-3} x \sec x \tan x dx$$

$$V = \tan x$$

$$\therefore \int_0^{\frac{\pi}{4}} \sec^n x dx = \tan x \sec^{n-2} x \Big|_0^{\frac{\pi}{4}} - (n-2) \int_0^{\frac{\pi}{4}} \sec^{n-2} x \tan^2 x dx$$

$$\therefore I_n = (\sec \frac{\pi}{4})^{n-2} - 0 - (n-2) \int_0^{\frac{\pi}{4}} \sec^{n-2} x (\sec^2 x - 1) dx$$

$$= (\sqrt{2})^{n-2} - (n-2) \int_0^{\frac{\pi}{4}} \sec^n x dx - \int_0^{\frac{\pi}{4}} \sec^{n-2} x dx$$

$$\therefore (n-2+1) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2}$$

$$\therefore I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{(n-2)}{(n-1)} I_{n-2}$$

$$[1] I_n$$

$$\therefore I_4 = \frac{\sqrt{2}^4}{5} + \frac{4}{5} I_2$$

$$= \frac{(\sqrt{2})^4}{5} + \frac{4}{5} \left[\frac{(\sqrt{2})^2}{3} + \frac{2}{3} I_0 \right] = \frac{4}{5} + \frac{4}{5} \left[\frac{2}{3} + \frac{2}{3} \left[\frac{\sqrt{2}^0}{1} + \frac{0}{1} I_0 \right] \right]$$

$$= \frac{4}{5} + \frac{4}{5} \left[\frac{2}{3} + \frac{2}{3} \right] = \frac{4}{5} + \frac{4}{5} \left[\frac{4}{3} \right]$$

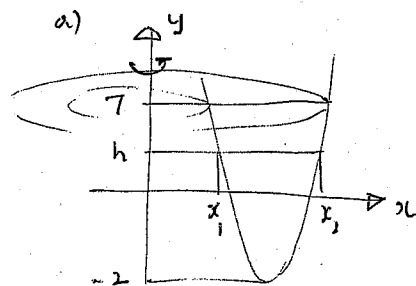
$$= \frac{4}{5} + \frac{16}{15}$$

$$= \frac{28}{15}$$

$$[1] \text{ Sub}$$

Question 6 [15 MARKS]

a)



$$y = (x-5)^2 - 2$$

$$h = (x-5)^2 - 2$$

$$\therefore x = 5 \pm \sqrt{h+2}$$

$$x_1 = 5 - \sqrt{h+2}$$

$$x_2 = 5 + \sqrt{h+2}$$

$$[1] x_1, x_2$$

$$(iii) A = \pi(R^2 - r^2) = \pi[(R+r)(R-r)]$$

$$= \pi[(10)(2\sqrt{h+2})]$$

$$[1] A(h)$$

$$(iii) \Delta V = A(h) \Delta h = 20\pi\sqrt{h+2} \Delta h$$

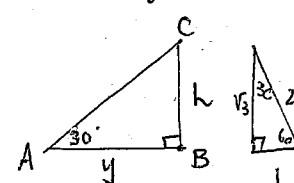
$$\therefore V = \lim_{\Delta h \rightarrow 0} \sum_{h=-2}^7 20\pi\sqrt{h+2} \Delta h = 20\pi \int_{-2}^7 \sqrt{h+2} dh$$

$$= 20\pi \cdot \frac{2}{3} [(h+2)^{\frac{3}{2}}]_{-2}^7 = \frac{40\pi}{3} [9^{\frac{3}{2}} - 0]$$

$$= \frac{27 \times 40\pi}{3} = 360\pi u^3$$

$$[1] \text{ Answer}$$

$$b) A = \frac{1}{2}bh$$



$$\tan 30 = \frac{h}{y}$$

$$h = y \tan 30 = \frac{y}{\sqrt{3}}$$

$$y = \sqrt{16-x^2}$$

$$(i) \therefore A = \frac{1}{2} y \cdot \frac{y}{\sqrt{3}} = \frac{1}{2\sqrt{3}} \sqrt{16-x^2} \cdot \sqrt{16-x^2}$$

$$\therefore A(x) = \frac{16-x^2}{2\sqrt{3}}$$

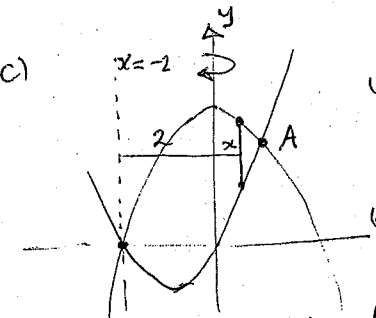
$$V = \int_{-4}^4 A(x) dx = \int_{-4}^4 \frac{16-x^2}{2\sqrt{3}} dx$$

$$= \frac{1}{\sqrt{3}} \int_{-4}^4 (16-x^2) dx = \frac{1}{\sqrt{3}} \left[16x - \frac{x^3}{3} \right]_{-4}^4$$

$$= \frac{128}{3\sqrt{3}} = \frac{128\sqrt{3}}{9} u^3$$

$$[1] \text{ Answer}$$

c)



$$(i) 4 - x^2 = x^2 + 2x \therefore 2x^2 + 2x - 4 = 0$$

$$\therefore 2(x^2 + x - 2) = 0 \quad 2(x-1)(x+2) = 0$$

$$\therefore x = 1, -2 \quad \therefore A(1, 3)$$

$$[1] A$$

$$(ii) \Delta V = 2\pi r h \Delta r$$

$$\Delta V = 2\pi (2+x) [4-x^2 - (x^2+2x)] \Delta x$$

$$= 4\pi (2+x) (2-x-x^2) \Delta x$$

$$= 4\pi [4-3x^2-x^3] \Delta x$$

$$(ii) \text{ cont. } \therefore V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2}^1 4\pi (4-3x^2-x^3) \Delta x = 4\pi \int_{-2}^1 (4-3x^2-x^3) dx$$

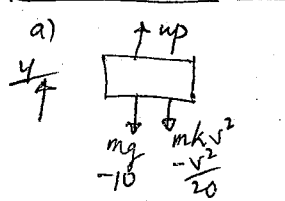
$$(iii) V = 4\pi \left[4x - x^3 - \frac{x^4}{4} \right]_{-2}^1 = \left[4 - 1 - \frac{1}{4} - [-8 + 8 - 4] \right]$$

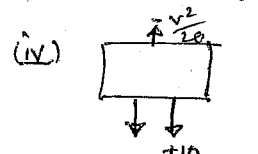
$$= 4\pi \cdot \left[2\frac{3}{4} + 4 \right] = 6\frac{3}{4} \cdot 4\pi = 27\pi u^3$$

$$[1] \text{ Eval. I}$$

$$[1] A$$

Question 7 [15 Marks]

- a)  $m\ddot{y} = -mg - \frac{mv^2}{20} \therefore \ddot{y} = -10 - \frac{v^2}{20}$
 $\therefore v \frac{dv}{dy} = \frac{-200 - v^2}{20} \therefore \frac{dv}{dy} = \frac{-200 - v^2}{20v}$
 $\therefore \frac{dy}{dv} = \frac{-20v}{200 + v^2} \therefore y \Big|_0^y = -10 \int_0^v \frac{2v}{200 + v^2} dv$
 $\therefore y = -10 \ln(200 + v^2) + 10 \ln 600 = 10 \ln \left(\frac{600}{200 + v^2} \right)$
 $\therefore -\frac{y}{10} = \ln \left(\frac{200 + v^2}{600} \right) \Rightarrow e^{-\frac{y}{10}} = \frac{200 + v^2}{600}$
 $\therefore 600e^{-\frac{y}{10}} = 200 + v^2 \Rightarrow v^2 = 600e^{-\frac{y}{10}} - 200$
 (ii) Max height $v=0 \therefore 200 = 600e^{-\frac{y}{10}} \Rightarrow \frac{1}{3} = e^{-\frac{y}{10}}$
 $\ln \frac{1}{3} = -\frac{y}{10} \therefore y = -10 \ln \frac{1}{3} = 10.99 \text{ m (10 lns)} \quad \square$
 (iii) $\frac{dv}{dt} = \frac{-v^2 - 200}{20} \therefore dt = \frac{-20 dv}{v^2 + 200} \quad dt \Big|_0^t = \frac{-20}{\sqrt{200}} \tan^{-1} \frac{v}{\sqrt{200}} \Big|_0^t$
 $\therefore t = \frac{-20}{\sqrt{200}} \tan^{-1} 0 + \frac{20}{\sqrt{200}} \tan^{-1} \left(\frac{20}{\sqrt{200}} \right) \approx 1.35 \text{ secs} \quad \square \text{ Answer}$

- (iv)  $\ddot{y} = 10 - \frac{v^2}{20} \therefore v \frac{dv}{dy} = \frac{200 - v^2}{20}$
 $\therefore \frac{dy}{dv} = \frac{20v}{200 - v^2} \quad dy \Big|_0^{10 \ln 3} = -10 \int_0^v \frac{-2v dv}{200 - v^2}$
 $\therefore 10 \ln 3 = -10 \ln(200 - v^2) \Big|_0^v = -10 \ln(200 - v^2) + 10 \ln 200$
 $\therefore \ln 3 = -\ln(200 - v^2) + \ln 200 \Rightarrow \ln \frac{3}{200} = -\ln(200 - v^2)$
 $\therefore \frac{200}{3} = 200 - v^2 \quad v^2 = 200 - \frac{200}{3} = \frac{400}{3} \therefore v = \frac{20}{\sqrt{3}} \approx 11.55 \text{ m/s}$

- b) (i) $V = Ah \therefore \frac{dV}{dt} = A \frac{dh}{dt}$ given $\frac{dV}{dt} = -k\sqrt{h} \quad \square \text{ Rate from } V = Ah$
 $A \frac{dh}{dt} = -k\sqrt{h} \Rightarrow \frac{dh}{dt} = -\frac{k}{A} \sqrt{h}$
 (ii) $\frac{dh}{dt} = -\frac{k}{A} h^{\frac{1}{2}}$ separate $h^{-\frac{1}{2}} dh = -\frac{k}{A} dt$ integrate $\square \text{ Separate}$
 $\int_{h_0}^h h^{-\frac{1}{2}} dh = \int_0^t -\frac{k}{A} dt \Rightarrow \frac{3}{2} h^{\frac{3}{2}} \Big|_{h_0}^h = -\frac{k}{A} t$
 $\therefore \frac{3}{2} [h^{\frac{3}{2}} - h_0^{\frac{3}{2}}] = -\frac{k}{A} t \therefore \frac{3}{2} h^{\frac{3}{2}} = \frac{3}{2} h_0^{\frac{3}{2}} - \frac{k}{A} t$
 $\therefore \frac{3}{2} h^{\frac{3}{2}} = \frac{3}{2} h_0^{\frac{3}{2}} - \frac{k}{A} t$
 $\therefore h^{\frac{3}{2}} = h_0^{\frac{3}{2}} - \frac{k}{\frac{3}{2}A} t \rightarrow h^{\frac{3}{2}} = h_0^{\frac{3}{2}} \left(1 - \frac{t}{T} \right) \therefore h^2 = h_0^2 \left(1 - \frac{t}{T} \right)^3 \quad \square$

- (iii) $h = \frac{h_0}{2} \quad t = 12 \text{ secs} \quad \left(\frac{h_0}{2} \right)^2 = h_0^2 \left(1 - \frac{12}{T} \right)^3 \therefore \frac{1}{4} = \left(1 - \frac{12}{T} \right)^3$
 $1 - \frac{12}{T} = \sqrt[3]{\frac{1}{4}} \therefore \frac{12}{T} = 1 - \sqrt[3]{\frac{1}{4}} \therefore T = \frac{12}{1 - \sqrt[3]{\frac{1}{4}}} \approx 32.428 \approx 32 \text{ secs (to nearest sec.)} \quad \square$

Question 8 [15 Marks]

- a) (i) $z + \frac{1}{z} = 2 \cos \alpha \therefore z^2 - 2 \cos \alpha z + 1 = 0 \quad \square \text{ quad.}$
 $\therefore (z - \cos \alpha)^2 - (\cos \alpha)^2 + 1 = 0 \therefore (z - \cos \alpha)^2 = -1 + \cos^2 \alpha$
 $\therefore z - \cos \alpha = \pm i \sin \alpha \therefore z = \cos \alpha \pm i \sin \alpha \quad \square z = \frac{\cos(\alpha)}{\cos(\alpha)}$
 $\therefore z = \cos \alpha \text{ or } \cos(-\alpha)$
 If $z = \cos \alpha$ then by de Moivre's theorem
 $z^n = \cos n\alpha$ and $z^{-n} = \cos(-n\alpha) \cos(-n\alpha) = \cos n\alpha$
 If $z = \cos \alpha \quad z^n + z^{-n} = \cos n\alpha + \cos n\alpha = 2 \cos n\alpha \quad \square \text{ result.}$
 or $z = \cos(-\alpha) = 2 \cos n\alpha$

- (ii) Let $w = z + \frac{1}{z} \quad w^2 = \left(z + \frac{1}{z} \right)^2 = z^2 + \frac{1}{z^2} + 2$
 Now $w^3 + w^2 - 2w - 2 = w^2(w+1) - 2(w+1) \quad \square$
 $= (w+1)(w^2 - 2) = \left(z + \frac{1}{z} + 1 \right) \left(z^2 + \frac{1}{z^2} \right)$
 $= z^3 + \frac{1}{z} + z + \frac{1}{z^3} + z^2 + \frac{1}{z^2}$
 $= z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3} \quad \square$

- (iii) $z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3} = 2 \cos \alpha + 2 \cos 2\alpha + 2 \cos 3\alpha = 0$
 $\therefore \cos \alpha + \cos 2\alpha + \cos 3\alpha = 0 \quad \square$
 $\therefore w = -1, \sqrt{2} \text{ or } -\sqrt{2} \therefore 2 \cos \alpha = -1, \sqrt{2} \text{ or } -\sqrt{2}$
 If $\cos \alpha = -\frac{1}{2}, \alpha = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \cos \alpha = \frac{1}{\sqrt{2}}, \alpha = \frac{\pi}{4} \text{ or } \frac{7\pi}{4} \quad \square$
 $\cos \alpha = -\frac{1}{\sqrt{2}}, \alpha = \frac{3\pi}{4}, \frac{5\pi}{4}$
 $\therefore \text{Six solutions } \alpha = \frac{\pi}{4}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{4\pi}{3} \text{ and } \frac{7\pi}{4} \quad \square$

- b) $\int_0^a f(x) dx = \int_a^0 f(a-u) du$ let $u = a - x, x=0 \quad u=a \quad du = -dx$
 $\therefore \int_0^a f(x) dx = \int_a^0 f(a-u) (-du) = \int_0^a f(u) du = \int_0^a f(x) dx = \text{LHS}$
 $\therefore \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{1 + \cos^2(\pi-x)} dx = \int_0^{\pi} \frac{(\pi-x) \sin x}{1 + \cos^2 x} dx \quad \square$
 $\cos(\pi-x) = -\cos x$
 $\cos^2(\pi-x) = \cos^2 x$
 $\sin(\pi-x) = \sin x$
 $\therefore I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^{\pi} \frac{(\pi-x) \sin x}{1 + \cos^2 x} dx$
 $\therefore 2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = -\pi \left[\tan^{-1}(\cos x) \right]_0^{\pi}$
 $\therefore 2I = -\pi \left[-\frac{\pi}{4} - \frac{\pi}{4} \right] = \frac{\pi^2}{2} \therefore I = \frac{\pi^2}{4} \quad \square$

- c) $x^2 + y^2 = r^2$
 $\therefore f(x) = \sqrt{r^2 - x^2} \therefore A = \int_{-r}^r 2\pi \sqrt{r^2 - x^2} \sqrt{1 + \left(\frac{-x}{\sqrt{r^2 - x^2}} \right)^2} dx \quad \square$
 $2x + 2y \frac{dy}{dx} = 0$
 $\therefore \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$
 $f'(x) = \frac{-x}{\sqrt{r^2 - x^2}} \quad \square$
 $A = \int_{-r}^r 2\pi \sqrt{r^2 - x^2} \sqrt{1 + \frac{x^2}{r^2 - x^2}} dx$
 $= \int_{-r}^r 2\pi \sqrt{r^2 - x^2} \frac{r}{\sqrt{r^2 - x^2}} dx = \int_{-r}^r 2\pi r dx = 2\pi r \Big|_{-r}^r = 4\pi r^2 \quad \square$